No.

Date

高数B上. HW4.

习题 2.8.3.

把 $[0,1]$ 分为 $[0,\frac{1}{n}]$, $[\frac{1}{n},\frac{2}{n}]$, ..., $[\frac{n-1}{n},1]$.

取 ξ_i 为 $0, \frac{1}{n}, \frac{2}{n}, \cdots, \frac{n-1}{n}$

那么, Riemann 和为 $\frac{1}{n}\sum_{i=0}^{n-1}(\frac{i}{n})^2 = \frac{1}{n^3}(1^2+2^2+\cdots+(n-1)^2)$

$= \frac{1}{n^3}\cdot\frac{1}{6}(n-1)n(2n-1) \to \frac{1}{3}$

$(n\to\infty)$.

5. (1) $x\in(0,\frac{\pi}{2})$ 时, $1<1+\sin x<2$.

故 $\frac{\pi}{2} = \int_0^{\pi/2} 1\,dx < \int_0^{\pi/2} 1+\sin x\,dx < \int_0^{\pi/2} 2\,dx = \pi$. □

6. (2) $x\in(0,\frac{\pi}{2})$ 时, $|x|>|\sin x|$. 故 $x^2>\sin^2 x$

故 $\int_0^{\pi/2} x^2\,dx > \int_0^{\pi/2}\sin^2 x\,dx$.

习题 2.9.1 (4) $L(x) = \int_x^{x^2} e^{-t^2}dt = \int_0^{x^2} e^{-t^2}dt - \int_0^x e^{-t^2}dt$.

$L'(x) = e^{-(x^2)^2}\cdot(x^2)' - e^{-x^2} = 2xe^{-x^4} - e^{-x^2}$.

5. 证明: $|F(x_1)-F(x_2)| = |\int_{x_1}^{x_2} f(t)\,dt|$ 假设 $x_2\ge x_1$

$\le \int_{x_1}^{x_2}|f(t)|dt \le \int_{x_1}^{x_2} L\,dt = L|x_1-x_2|$ □.

6. $G(x) = \int_0^x e^t\int_0^t \sin z\,dz\,dt$. 记 $g(t) = e^t\int_0^t \sin z\,dz\,dt$

$= \int_0^x g(t)\,dt$

$\Rightarrow G'(x) = g(x) = e^x\int_0^x \sin z\,dz$.

$= e^x f(x)$

记 $f(x) = \int_0^x \sin z\,dz$

$f'(x) = \sin x$

$\Rightarrow G''(x) = e^x(f(x)+f'(x))$

$= e^x(\int_0^x \sin z\,dz + \sin x)$ ~~$= e^x(\sin x+\cos x)$~~.

$= e^x(1-\cos x+\sin x)$.

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习题 2.10.1.(5) $\int_0^{\pi} (2\sin x + x^3)\,dx = \left(\frac{x^4}{4} - 2\cos x\right)\Big|_0^{\pi} = \frac{\pi^4}{4} + 4.$

(6) $\int_0^1 x^5 + \frac{1}{3}x^3 + \frac{1}{2}x + 1\,dx = \left(\frac{x^6}{6} + \frac{1}{12}x^4 + \frac{1}{4}x^2 + x\right)\Big|_0^1$

$= \frac{1}{6} + \frac{1}{12} + \frac{1}{4} + 1 = \frac{3}{2}.$

3. (1) $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{n} \sin\frac{k}{n} = \int_0^1 \sin x\,dx = 1 - \cos 1$

(2) $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{k^3}{n^4} = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^3 = \int_0^1 x^3\,dx = \frac{1}{4}.$

4. (3). $\int_0^1 x\left|\frac{1}{2} - x\right|\,dx = \int_0^{\frac{1}{2}} x\left(\frac{1}{2} - x\right)dx + \int_{\frac{1}{2}}^1 x\left(x - \frac{1}{2}\right)dx$ + C

$= \int_0^{\frac{1}{2}} \left(\frac{x}{2} - x^2\right)dx + \int_{\frac{1}{2}}^1 \left(x^2 - \frac{x}{2}\right)dx$

$= \left(\frac{x^2}{4} - \frac{x^3}{3}\right)\Big|_0^{\frac{1}{2}} + \left(\frac{1}{3}x^3 - \frac{x^2}{4}\right)\Big|_{\frac{1}{2}}^1 = \frac{1}{8}.$

(4) $\int_0^{2\pi} |\sin x|\,dx = 2\int_0^{\pi} \sin x\,dx$ (周期 π).

$= 2(-\cos x)\Big|_0^{\pi} = 4.$

(5) $\int_0^2 (x - [x])\,dx = 2\int_0^1 (x - [x])\,dx$ (周期 1)

$= 2\int_0^1 x\,dx$ ($x \in (0,1)$, $[x] = 0$)

$= 1.$

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